

# STA 013 — Problem Set

Topic: Intro to Statistics, Variables & Graphing | Based on Lecture: June 22 (Professor Max)

---

**Instructions:** Choose the best answer for each multiple-choice question. For Question 12, select the answer that best reflects sound statistical reasoning.

**1. Which statement correctly distinguishes a *population* from a *sample*?**

- A) A population is always larger than 1,000 units, while a sample is always smaller.
- B) A sample is the entire group of interest; a population is a subset drawn from it.
- C) A population is the entire group of interest; a sample is a subset drawn from the population that is used to make inferences about it.
- D) Populations only apply to people, while samples can apply to any object.

**2. A researcher wants to study the average commute time of all employees at a company with 10,000 workers. She randomly surveys 75 employees. What is the *experimental unit* in this study?**

- A) The company
- B) The number 75
- C) One employee
- D) The commute time itself

**3. Suppose a data set records both the height and weight of every student in a class. This is an example of:**

- A) Univariate data
- B) Bivariate data
- C) Multivariate data
- D) Qualitative data

**4. Which of the following is an example of a *discrete* quantitative variable?**

- A) The exact weight of a newborn baby in kg
- B) The time it takes to finish a 5K race
- C) The number of siblings a person has
- D) The yield, in kg, of corn from a 2-hectare field

**5. A survey asks respondents to rate a new policy as “Strongly Disagree,” “Disagree,” “Neutral,” “Agree,” or “Strongly Agree.” Even though these responses could be coded as numbers (1–5) for analysis, this variable is best classified as:**

- A) Discrete quantitative
- B) Continuous quantitative
- C) Qualitative

D) Bivariate

→ *Tricky concept check: just because a variable can be coded numerically does not make it quantitative — consider what is fundamentally being measured.*

**6. Fill in the blank: A \_\_\_\_\_ results when a variable is actually measured on an experimental unit, while a set of these is called \_\_\_\_\_.**

- A) sample; a population
- B) measurement; data
- C) population; a sample
- D) variable; a distribution

**7. A city's transportation department records the number of traffic accidents per month along a specific highway for the past 12 months. This single quantitative variable, tracked across time, should be best displayed using a:**

- A) Pie chart
- B) Bar chart comparing categories
- C) Time series (line) chart
- D) Histogram only, never a line chart

**8. A data set lists the cause of death (e.g., "heart disease," "cancer," "accident," "other") for 38 deceased U.S. presidents. This variable is:**

- A) Discrete quantitative
- B) Continuous quantitative
- C) Qualitative
- D) Bivariate

**9. In a relative frequency table, the relative frequencies for four categories are found to be 0.15, 0.35, 0.10, and 0.30. Without knowing anything else, what can you conclude?**

- A) The data set is invalid because relative frequencies must be whole numbers.
- B) There must be a fifth category with relative frequency 0.10, since all relative frequencies must sum to 1.
- C) This data set must represent a population, not a sample.
- D) Nothing can be concluded; relative frequencies don't need to sum to anything specific.

**10. A grocery store wants to compare the average price of a dozen eggs across five different countries (one bar per country, bar height = average price). Even though "price" is a quantitative variable, this is typically graphed with a bar or pie chart because:**

- A) Quantitative variables can never use line graphs
- B) The x-axis represents categories (countries), not a continuous time scale
- C) Prices are technically qualitative once converted to currency

D) Bar charts are only used for population data, never samples

**11. A statistics class collects the eye color of every cat that visits the UC Davis campus veterinary clinic over one month. What is the *variable*, and what is the *experimental unit*?**

- A) Variable: the clinic; Experimental unit: one month
- B) Variable: eye color; Experimental unit: one cat (at UC Davis)
- C) Variable: one cat; Experimental unit: eye color
- D) Variable: UC Davis; Experimental unit: eye color

**12. True or False: “If a data set covers only the most recent 38 of 100 total historical records, it can always be safely treated as a representative sample of the full 100 records.”**

- A) True — any subset automatically counts as a representative sample.
- B) False — selecting only the most recent cases is not necessarily representative of the entire historical population, even though it could be analyzed as a population in its own right (e.g., “the population of the last 38 records”).
- C) True, but only if the subset is larger than 30 records (rule of 30).
- D) False — subsets can never be analyzed statistically under any circumstance.

## Answer Key & Explanations

---

1. **C** — Population = entire group of interest; sample = subset used to infer about the population.
2. **C** — The experimental unit is the individual object measured on — one employee.
3. **B** — Two variables (height and weight) measured on one experimental unit = bivariate.
4. **C** — Number of siblings has a natural counting gap (0, 1, 2, 3...) — discrete. (A, B, D are continuous.)
5. **C** — Despite being numerically coded, an ordinal rating like this represents a label/category, not a true numerical quantity — qualitative (mirrors the “politician rating” trick from lecture).
6. **B** — Measurement = value collected on one unit; data = the full set of measurements.
7. **C** — A single quantitative variable tracked over time = time series → line or bar chart, per lecture definition.
8. **C** — Cause of death is a label/category, not a number — qualitative.
9. **B** — All relative frequencies in a complete distribution must sum to exactly 1;  $0.15+0.35+0.10+0.30 = 0.90$ , leaving 0.10 unaccounted for.
10. **B** — Bar/pie charts work whenever the x-axis is categorical, regardless of whether the measured variable itself is qualitative or quantitative.
11. **B** — Variable = eye color (what changes); Experimental unit = one cat at UC Davis (what it's measured on).
12. **B** — Recency-based subsets are not automatically representative; whether it's a “sample” or its own “population” depends on how the population of interest is defined — directly echoing the presidents' age-at-death discussion in lecture.